

Same DSM objective, different score parameterizations

Continuous VE: shared backbone architecture and DSM objective; fixed training-data statistics

$$x \sim p_{\text{train}}, \quad t \sim \mathcal{U}[t_{\min}, 1], \quad \varepsilon \sim \mathcal{N}(0, I), \quad y = x + \sigma_t \varepsilon$$

Score-SDE baseline

Network predicts the full scaled score

$$\sigma_t s_{\theta}(y, t) = h_{\theta}(y, t)$$

$$\mathcal{L}_{\text{base}} = \mathbb{E} \|h_{\theta} + \varepsilon\|^2$$

Fourier Gaussian (ours)

Fixed Gaussian score + learned residual

$$\sigma_t s_{\theta} = \sigma_t s_G + \mathcal{F}^{-1}[b_t \mathcal{F} h_{\theta}]$$

μ, P_k : fixed training-only mean and spectrum

$$\mathcal{L}_{\text{FG}} = \mathbb{E} \|\sigma_t s_G + \mathcal{F}^{-1}[b_t \mathcal{F} h_{\theta}] + \varepsilon\|^2$$

Both optimize $\mathcal{L}_{\text{DSM}} = \mathbb{E} \|\sigma_t s_{\theta}(y, t) + \varepsilon\|^2$

What is fixed in the proposed adapter?

$$\sigma_t \hat{s}_{G,k} = -\sigma_t (\hat{y}_k - \hat{\mu}_k) / (P_k + \sigma_t^2)$$

$$b_{t,k} = \sqrt{P_k / (P_k + \sigma_t^2)}$$

The Gaussian reference need not equal the data distribution.

In residual coordinates, the loss retains $b_{t,k}^2$ weighting.

